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SARDAR PATEL UNIVERSITY

BCA (Semester-V) Examination – October 2025

[US05CBCA51] : Object Oriented Programming - III

Date: 04/10/2025 (Saturday)

Time: 10.00am to 1.00pm

Total Marks: 70

Note: Figure indicate at right is maximum marks for question.

Q -1 Select an appropriate option for the following statements. [10]

1. _____ package must import before use of JDBC in program.
[a] java.awt.* [b] java.util.*
[c] java.sql.* [d] java.net.*
2. The JDBC _____ class defines objects which can connect Java applications to a JDBC driver.
[a] Statement [b] DriverManager
[c] ResultSet [d] SQLManager
3. _____ is a java API which enables the java programs to execute SQL statements.
[a] JDBC [b] JDK
[c] JRE [d] Applet
4. The life cycle of a servlet is managed by _____.
[a] Servlet Context [b] Servlet Container
[c] The Supporting Protocol [d] All of these
5. What is javax.servlet.Servlet?
[a] Concrete class [b] Abstract class
[c] Interface [d] None of these
6. Which of the following code can be used to clear any data that exists in the buffer as well as the status code and headers?
[a] request.reset() [b] response.reset()
[c] response.resetBuffer() [d] None of these
7. Which is not a directive?
[a] include [b] page
[c] export [d] taglib
8. TLD Stands For _____.
[a] Tag Library Data [b] Tag Library Domain
[c] Tag Library Descriptor [d] Tag Library Description
9. The life cycle of thread in java is controlled by _____.
[a] JVM [b] JDK
[c] JRE [d] None of these
10. Which is the name of the method used to start a thread execution?
[a] run() [b] start()
[c] wait() [d] terminate()

Q-2 Answer the following questions in brief. (Attempt any Ten)

[20]

1. Define JDBC.
2. What is autoboxing?
3. Differentiate executeQuery() and executeUpdate().
4. Write use of HttpSession Interface.
5. Draw web-based architecture.
6. Briefly write about service() method of servlet.
7. Write the phases of JSP life cycle page?
8. What is JSP Directives?
9. Draw the structure of JSP Life Cycle.
10. Write about Thread scheduling.
11. List Java Thread methods.
12. Explain void start().

Q-3

[A] Write a note on JDBC architecture.

[05]

[B] Discuss types of JDBC Drivers

[05]

OR

[A] Explain Java Database Connectivity with different components.

[05]

[B] Write a steps of JDBC program for executing any query?

[05]

Q-4 Explain Servlet Life Cycle through diagram.

[10]

OR

Q-4 Discuss doGet() and doPost() method of HTTP Servlet Class using example.

[10]

Q-5

[A] Explain the execution process of JSP in detail.

[05]

[B] Explain attributes of JSP page directives

[05]

OR

Q-5

[A] Write a note on JSP components.

[05]

[B] Discuss scriptlet tag of JSP with example.

[05]

Q-6

[A] What is Thread? Discuss different ways to create a thread in Java.

[05]

[B] Write a note on Thread Life Cycle.

[05]

OR

[A] Discuss different ways to Implement Java Thread

[05]

[B] What is Thread Synchronization? Explain how to implement of synchronization using example. **[05]**
